1

a) transition function δ:

$$Q \times \Sigma \to Q : \begin{cases} \delta(q_0, j) = q_j \\ \delta(q_1, j) = q_{2(1-j)} \\ \delta(q_2, j) = q_{1+j} \end{cases}, \text{for each } j \in \{0,1\}$$

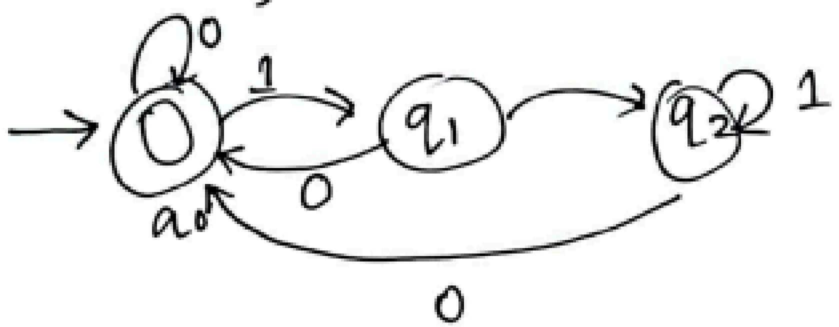
b) Given,

$Q = \{q_0, q_1, q_2\}$

$\Sigma = \{0, 1\}$

$q_{0*} = q_0$ [start state]

$F = \{q_0\}$



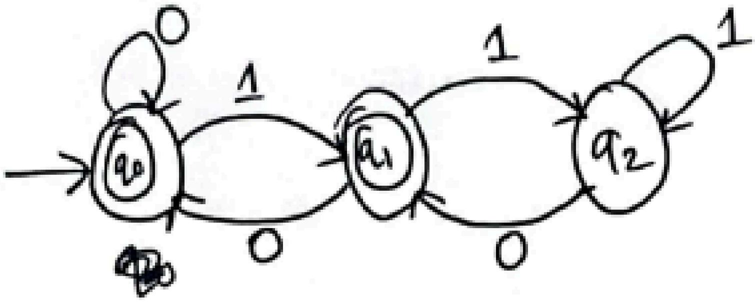
transition function δ:

$$Q \times \Sigma \to Q : \begin{cases} \delta(q_0, j) = q_j \\ \delta(q_1, j) = q_{2j} \\ \delta(q_2, j) = q_{2j} \end{cases}, \text{for each } j \in \{0,1\}$$

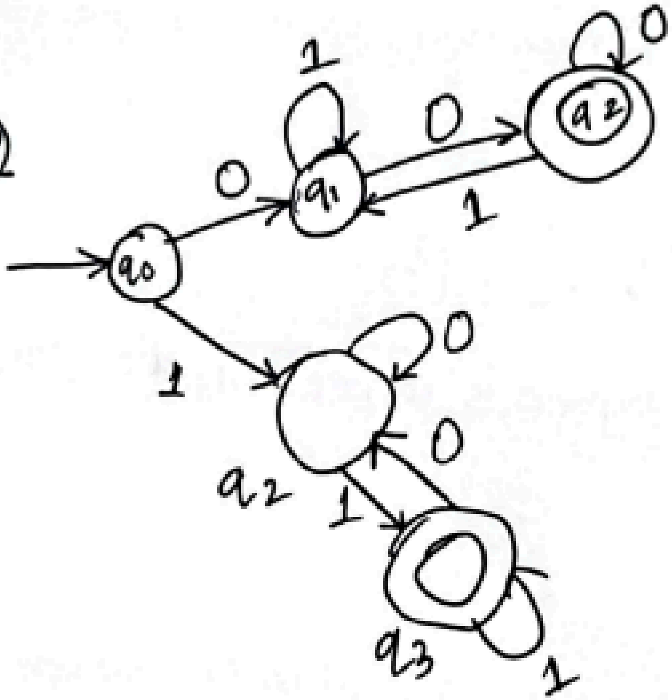
c) $L = \{ w \mid w$ represents an int divisible by 3$\}$

2

a)

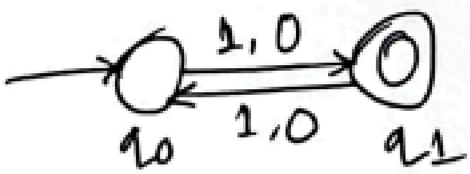


b)

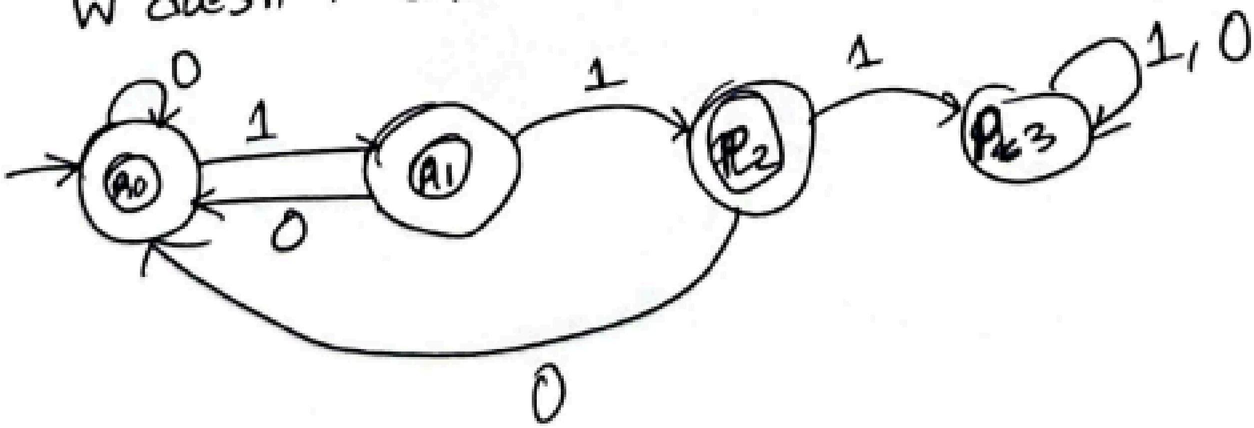


c)

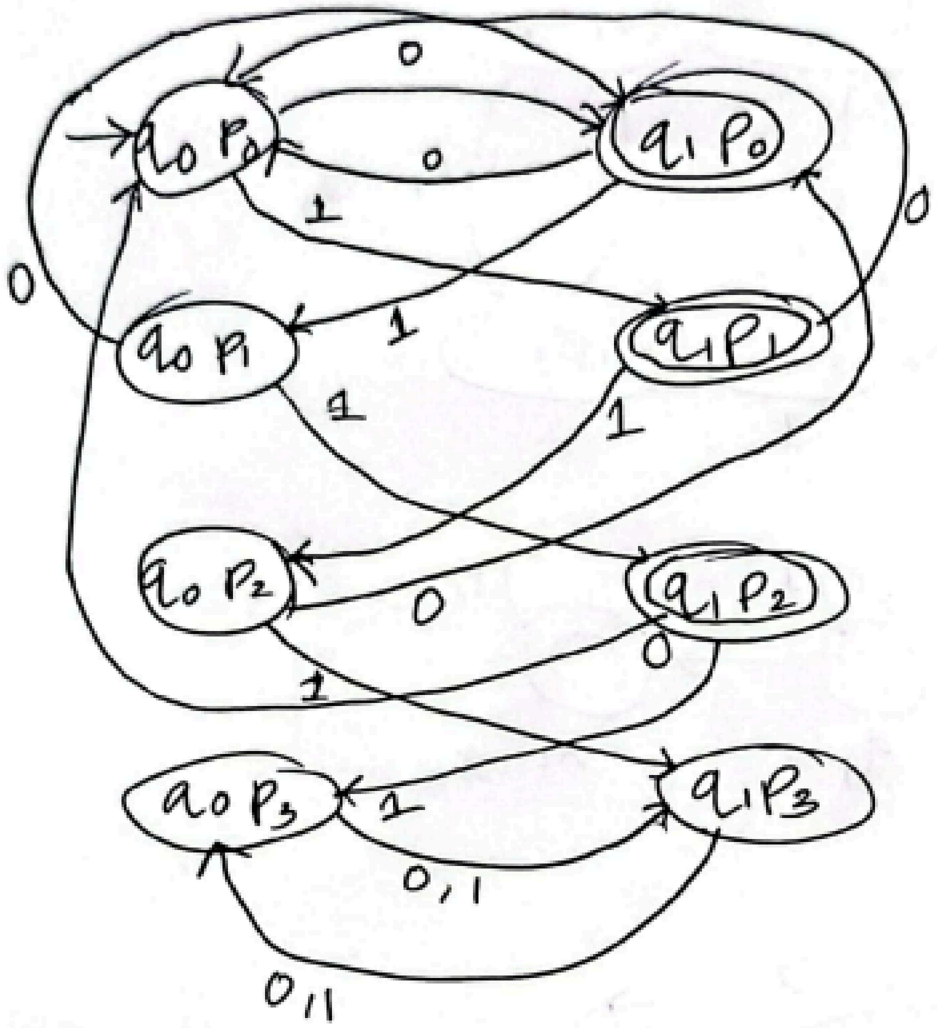
w doesn't have even length



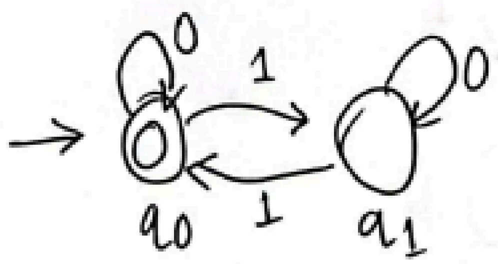
w doesn't contain the string 111



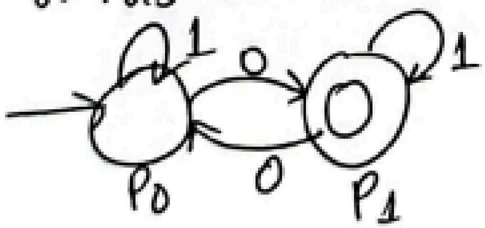
So, w doesn't have even length and w doesn't contain the string 111 will be:



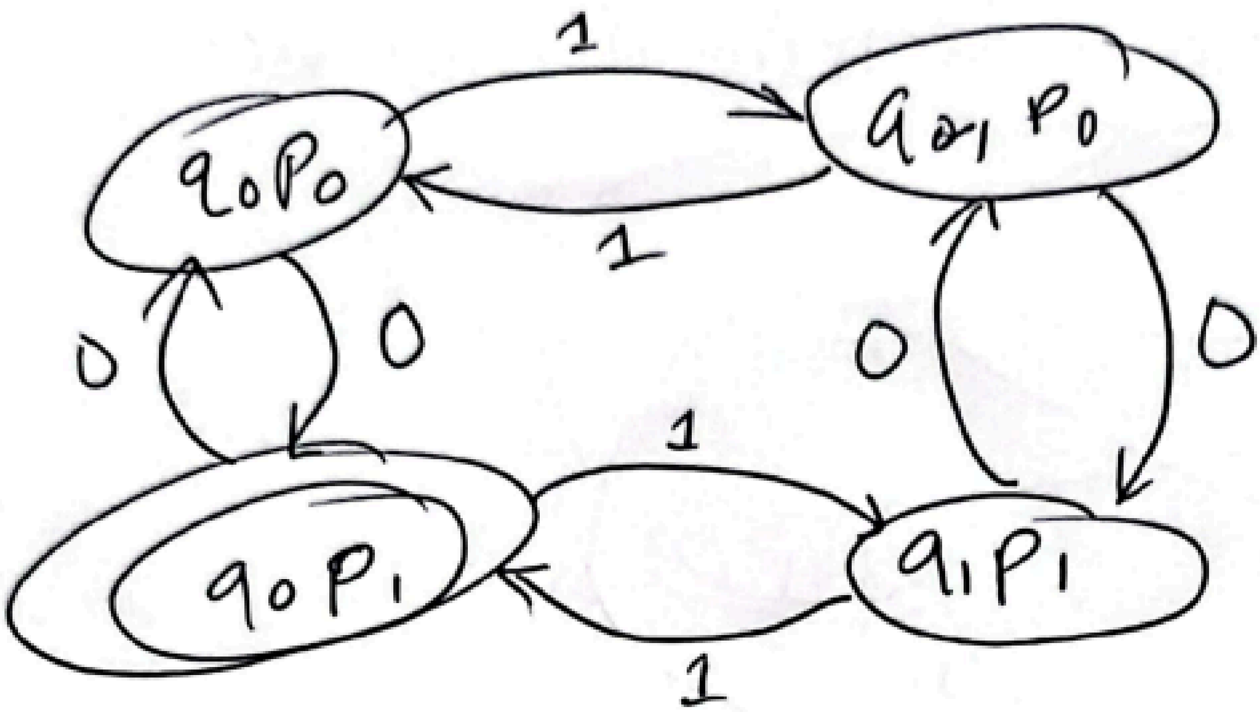
d) w doesn't have odd number 1's



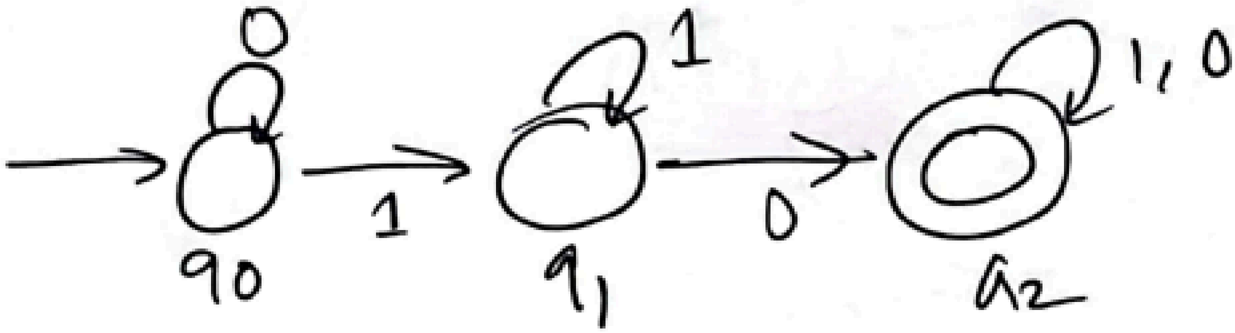
w has odd number 0's,



So, the DFA that doesn't have odd number 1's and has odd number 0's will be:

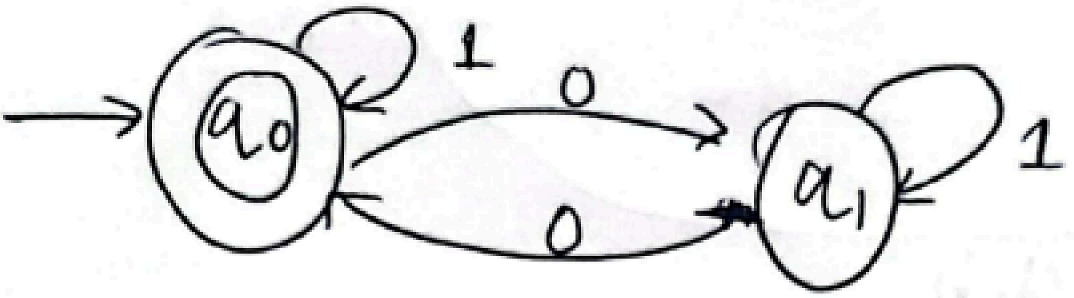


a)

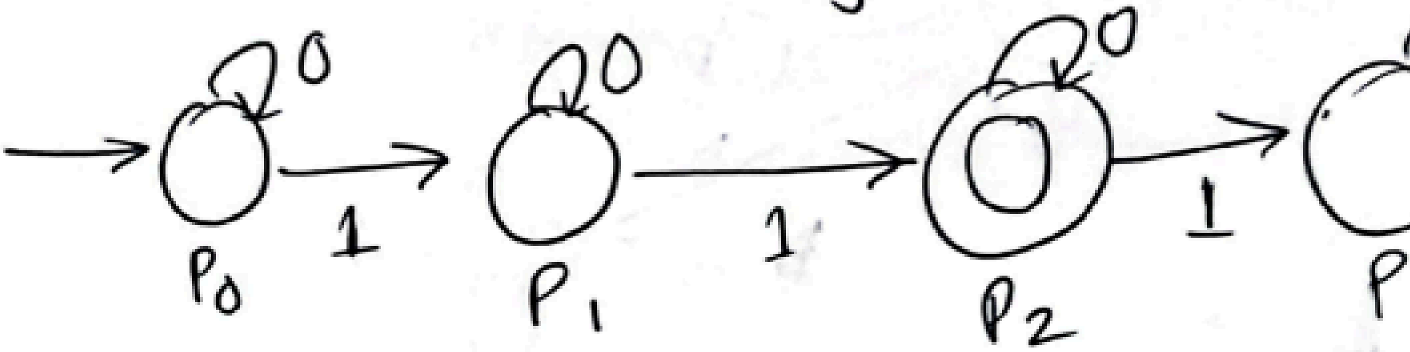


f) Let's assume

q = w has even number 0's



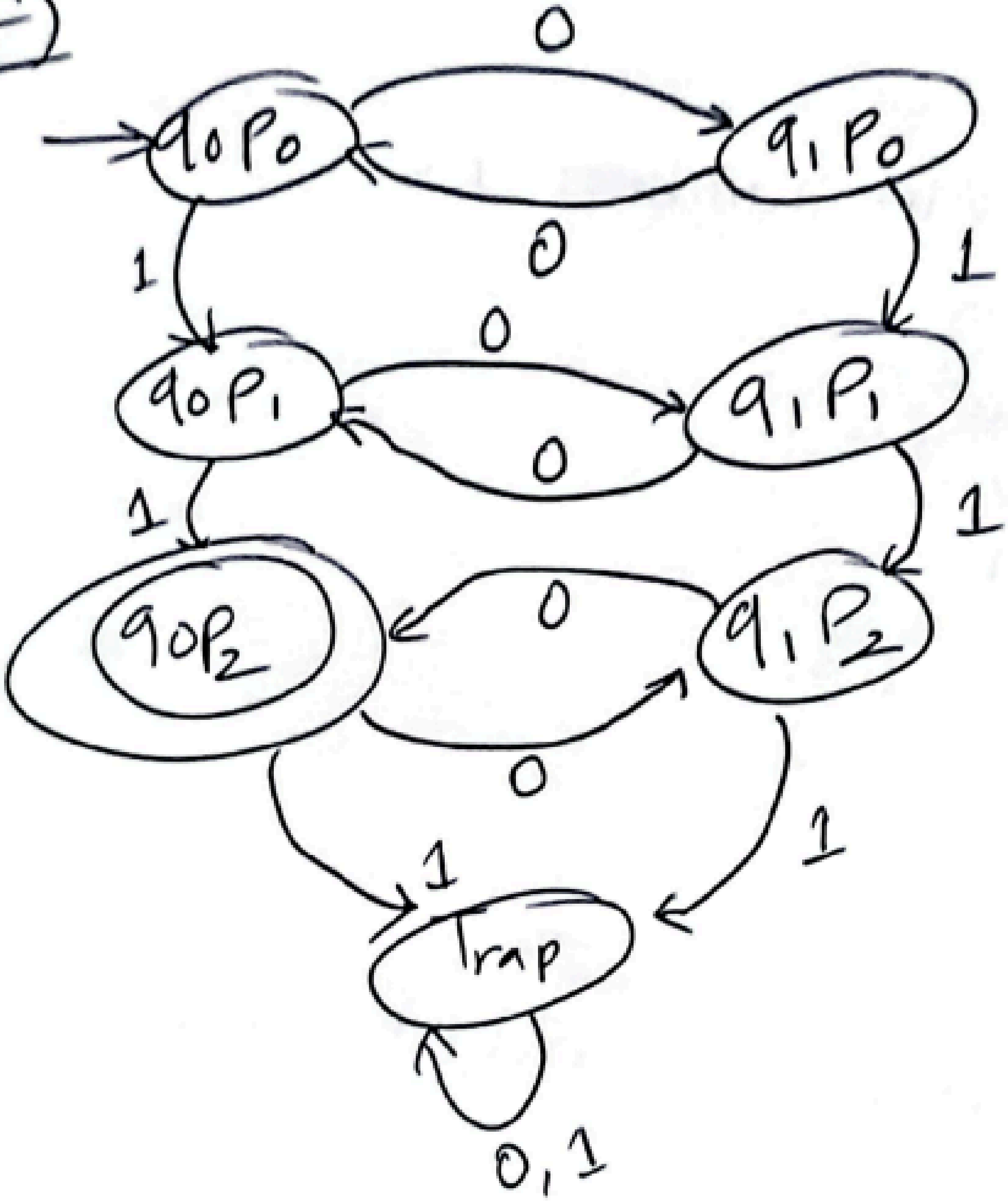
p = w has exactly two 1's



Now,

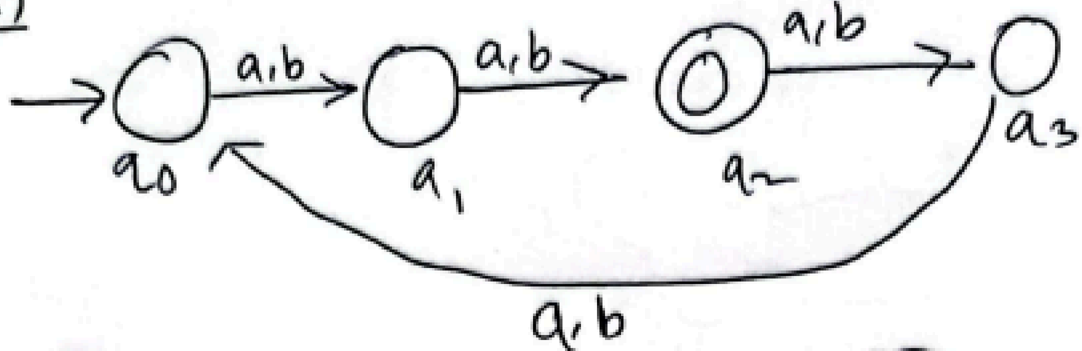
q ∨ p will be DFA.

f)

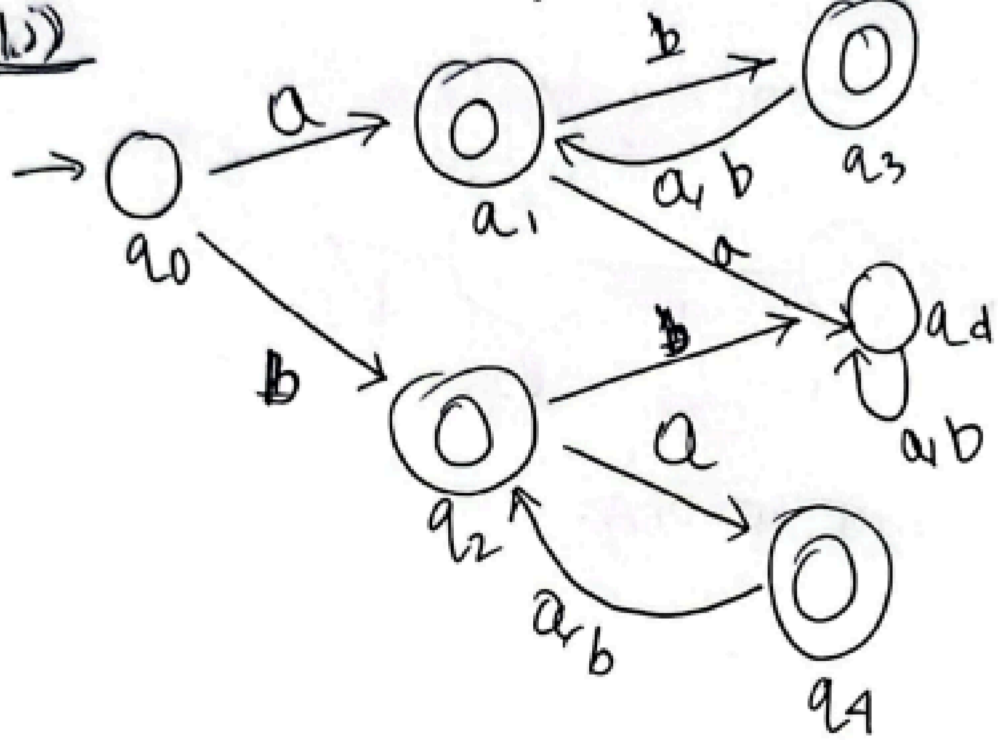


3

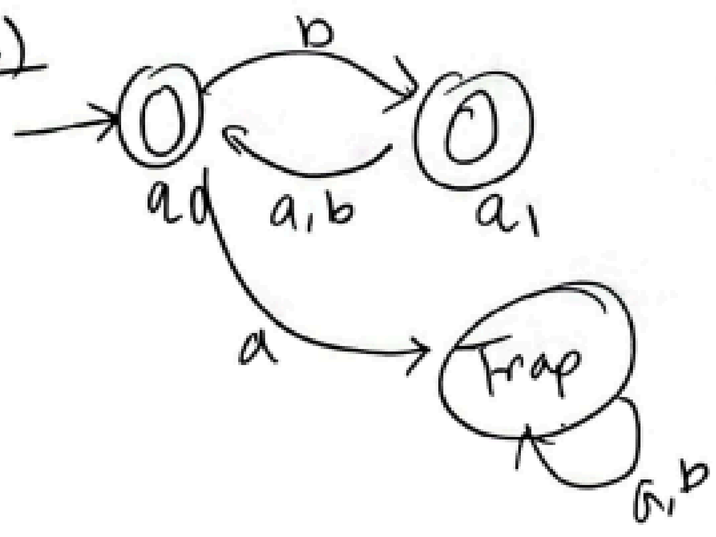
a)


a,b
a,b
a,b
q0
q1
q2
q3
a,b

b)


a
b
a,b
a
b
q0
q1
q3
qd
a,b
b
a
q2
a,b
q4

c)


b
q0
a,b
q1
a
Trap
a,b